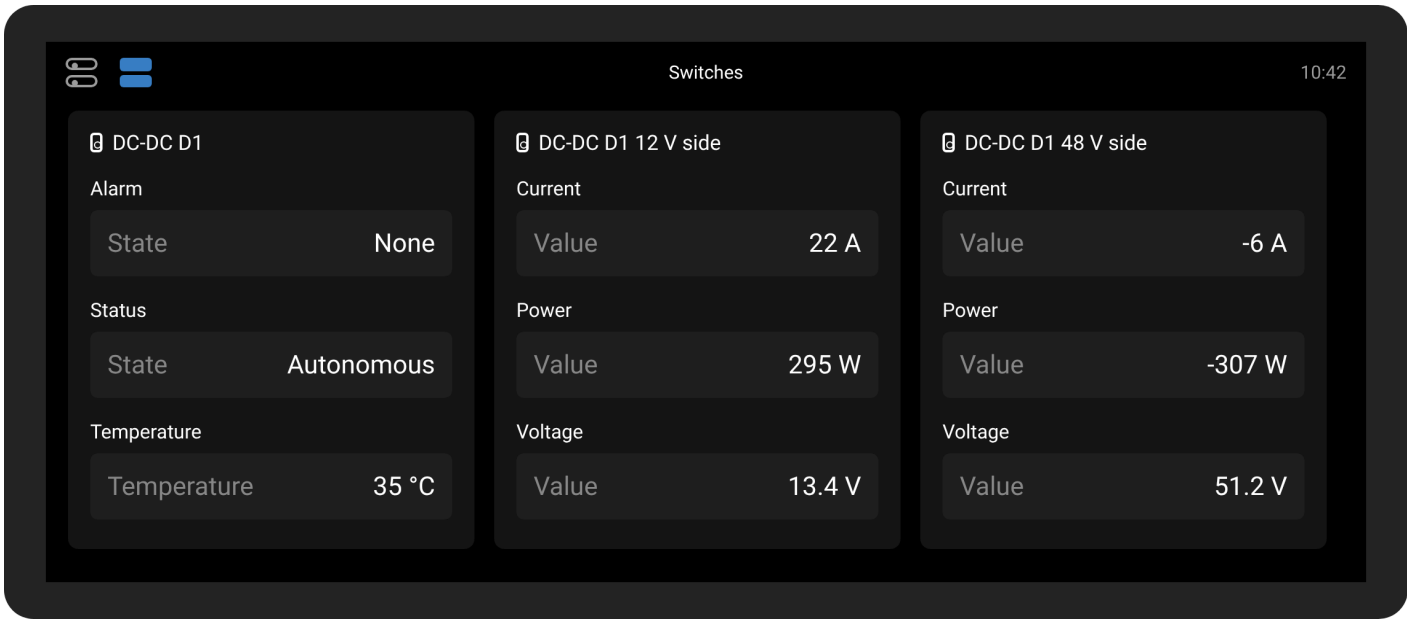
INSTALLATION GUIDE · MONITOR V2.0

# Nomadic D1 DC-DC converter monitor on your Victron Cerbo GX

Make the cable, set up the Cerbo, and install the monitor. When you're done, the D1's readings appear on the Cerbo screen, in the VRM app and on a web page with live charts.

**It only watches.** The monitor listens to the D1 and never sends it anything, so the D1 keeps working exactly as it does today, on its own built-in logic.




What you'll see on the Cerbo screen when you're done (drawn from Victron's own screen design, so the real screen may differ slightly).

### You'll be able to see

- 12 V and 48 V side voltage, current and power
- The D1's temperature, what it's doing (standby, autonomous, derating…) and every alarm, in plain words
- Charts of the last 24 hours on a web page
- Everything in the VRM app on your phone

### Good to know

- It takes about 1 hour, including making the cable
- Nothing changes for the D1. The Cerbo can't send it any commands
- You don't need to type any commands. The **Cerbo Setup** helper does it for you

**STEP 1** Get what you need

**STEP 2** Make the cable

**STEP 3** Set up the Cerbo

**STEP 4** Connect everything

**STEP 5** Install the monitor

**STEP 6** Use it

## 1 Get what you need

Gather everything before you start.

### For the cable

- A standard Ethernet patch cable (Cat5e or Cat6), long enough to reach from the Cerbo to the D1
- Two crimp contacts to add CAN_H and CAN_L to the D1's 6-pin signal plug (TE Connectivity connector 1-967616-1, per the D1 manual). Ask Nomadic or a TE distributor for the matching plug and contacts, or for a ready-made pigtail
- A ring terminal (M8) or a free spot on your ground bus, for the CAN ground wire
- Wire cutters, a wire stripper and a crimp tool for the TE contacts
- Heat-shrink tubing or electrical tape
- A multimeter with a continuity (beep) setting and an Ohms setting
- A VE.Can terminator: the small RJ45 plug that came in the Cerbo box
- A 120 Ω resistor (¼ W) for the D1 end. Required: the D1 has no built-in terminator

### For the software

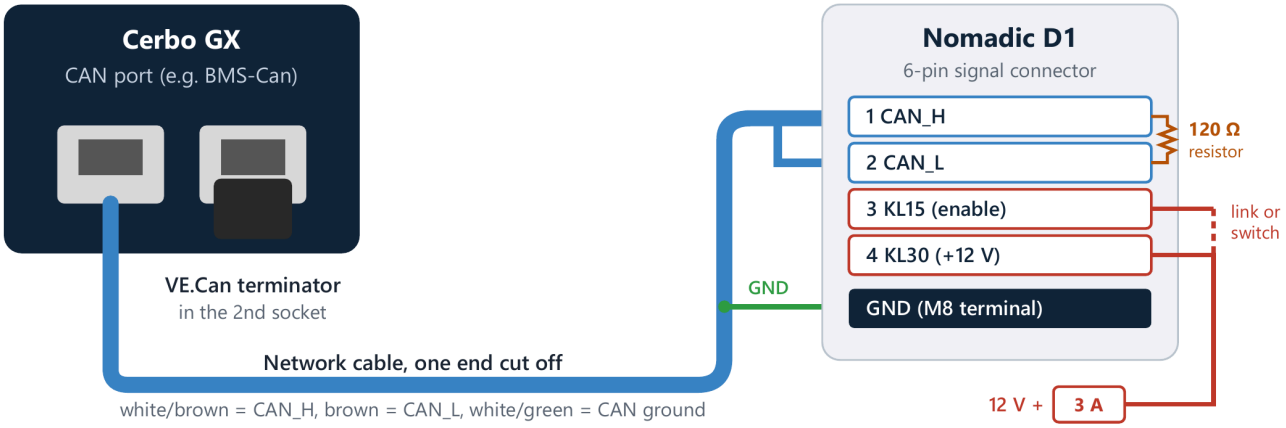
- A Windows PC on the same network (Wi-Fi or Ethernet) as the Cerbo
- The project folder downloaded from GitHub, which contains **Cerbo Setup.bat** and the **dbus-nomadic-dcdc** folder
- The Cerbo running Venus OS **v3.80 or newer**. Step 3 shows you how to check

> **Safety**
>
> Before you work on the D1's wiring, switch off its 12 V supply (KL30/KL15) and follow the D1 manual. The CAN cable only carries data, but it ends next to the D1's 48 V and 12 V power terminals.

## 2 Make the cable

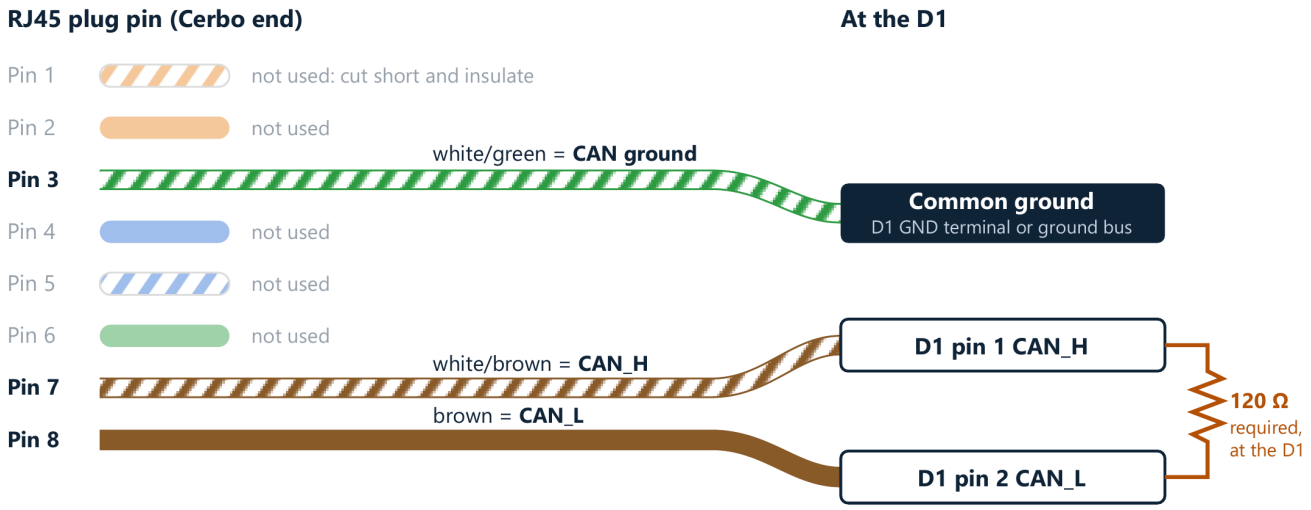
The Cerbo uses **RJ45 sockets**, the same plug as network cables. The easiest way is to take a ready-made network cable, keep the plug on one end for the Cerbo, and cut off the other end. You only need **3 of its 8 wires**: CAN_H and CAN_L go into the D1's signal plug, and the CAN ground goes to the common ground.

### The whole connection

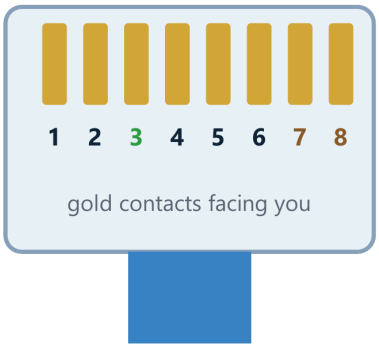



Blue: the network cable you make (CAN), with the required 120 Ω resistor across CAN_H and CAN_L at the D1. Green: CAN ground to the D1's common ground. Red: the D1's own 12 V wires (KL30 and KL15). If your D1 already works today, the red wires are already in place.

### Which wire goes where

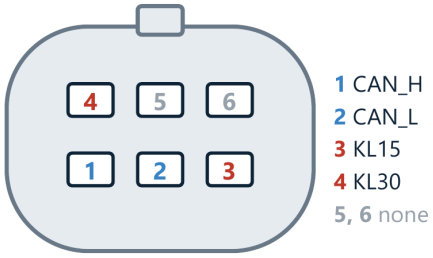



Pins 7 and 8 are the **brown pair** (twisted together), which suits CAN well. Pin 3 is **white/green** on most cables (T568B), or **white/orange** on T568A cables. The multimeter check in step 2b tells you which.




cable pointing down, clip at the back

**RJ45 plug (Cerbo end)**: pin 1 is on the left.




layout as drawn in the D1 manual
(facing the D1's connector)

**D1 6-pin signal connector**. Check the numbers moulded on the connector before you crimp.

| Wire | RJ45 pin (Cerbo) | Goes to (D1) |
|---|---|---|
| white/brown | 7 | **Pin 1 CAN_H** on the 6-pin signal connector |
| brown | 8 | **Pin 2 CAN_L** on the 6-pin signal connector |
| white/green (white/orange on T568A) | 3 | **Common ground**: the D1's GND terminal, or the ground bus it's bolted to. The D1's signal connector has no ground pin |
| The other 5 wires (pins 1, 2, 4, 5, 6) | | Not used. Cut them short and insulate each one |

Sources: Victron's *VE.Can to CAN-bus BMS cables manual* (Cerbo side: pin 7 CAN-H, pin 8 CAN-L, pin 3 GND) and the *D1 Bidirectional DC-DC Converter Manual*, Rev C (signal connector pinout).

## Make it

**2a** **Cut off one plug.** Keep the other plug for the Cerbo. Strip about 5 cm (2 in) of the grey outer jacket.

**2b** **Check the wires with a multimeter** on the continuity (beep) setting. Touch one probe to a gold contact on the plug and the other to a wire:

- Pin 7 should beep with **white/brown**
- Pin 8 should beep with **brown**
- Pin 3 should beep with **white/green**. If it beeps with white/orange instead, use white/orange as the CAN ground

**2c** **Insulate the 5 wires you don't need.** Cut them short and cover **each one** separately with heat-shrink or tape, so none can touch anything.

**2d** **Add CAN_H and CAN_L to the D1's signal plug.** With the D1 switched off, crimp contacts onto **white/brown** (into **pin 1, CAN_H**) and **brown** (into **pin 2, CAN_L**). Keep the brown pair twisted together as close to the plug as you can. If your D1 is already wired, unplug the existing signal plug and add the two contacts to it.

**2e** **Connect the CAN ground.** Put a ring terminal on the **white/green** wire and connect it to the **common ground**: the D1's GND terminal, or the ground bus that terminal is bolted to.

> Never connect any of the network-cable wires to the D1's HV+ or LV+ terminals. CAN_H and CAN_L only go to pins 1 and 2 of the signal plug.

**2f** **Check the D1's own 12 V wires (KL30 and KL15).** The D1 needs both to work at all, and without them it's also silent on CAN. If your D1 already works today, they're already in place: just check them.

| Pin | Connect to |
|---|---|
| 4<br>KL30 | A permanent 12 V supply (9–16 V, up to 2 A) through a 3 A fuse close to the 12 V source |
| 3 KL15 | A 12 V "enable" signal (8 V or more = on). Link it to KL30 to keep the D1 always enabled, or wire it through a switch, ignition or relay |
| 5, 6 | Nothing. The manual says "do not connect" |

**2g** **Fit the 120 Ω resistor at the D1.** The D1 has no built-in terminator, so connect a 120 Ω resistor between CAN_H and CAN_L at the D1 end, for example between the white/brown and brown wires where they meet the signal plug. Insulate it with heat-shrink.
**Check:** with the RJ45 plug not yet in the Cerbo, measure between the white/brown and brown wires (or pins 7 and 8 of the RJ45 plug). It should read about **120 Ω**.

**2h** **Final check:** plug the RJ45 into the Cerbo port you'll use, with the VE.Can terminator in that port's other socket. With everything switched off, measure between white/brown and brown again:

| You measure | Meaning |
|---|---|
| About 60 Ω | Correct: both ends are terminated |
| About 120 Ω | The VE.Can terminator isn't in the Cerbo port's other socket, or the cable is in a different port |
| Very high or "OL" | The resistor at the D1 isn't connected. Re-check step 2g |

## 3 Set up the Cerbo

Do this on the Cerbo's touchscreen, or remotely with Remote Console in VRM. Every item below is found from the Settings menu.

**3a** **Check the firmware version.** Go to **Settings › General › Firmware**. It needs to be **v3.80 or newer**. If it's older, use **Online updates** on the same page to update first.

**3b** **Write down the Cerbo's IP address.** Go to **Settings › Connectivity › Ethernet** (or Wi-Fi) and read the IP address.
IP address: _____

**3c** **Choose the CAN port and speed.** Go to **Settings › Connectivity › BMS-Can port** (or another CAN port that has nothing else plugged in) and set **CAN-bus profile** to **Up, but no services (250 kbit/s)**. The D1 talks at either 250 or 500 kbit/s. Start with 250. Step 5c shows each port's speed, and tells you if the D1 is talking at the other speed. Then change the profile to **Up, but no services (500 kbit/s)**.

**3d** **Let your PC connect to the Cerbo** (this switches on SSH):

1. Go to **Settings › General › Access & Security › Access level** and choose **User & Installer**. The password is `ZZZ`.
2. Still on the Access & Security page, drag the list down with your finger and hold it for about 5 seconds. The access level changes to **Superuser**.
3. Tap **Root password** and choose a password (at least 8 characters).
   Root password: _____
4. Switch **Enable SSH on LAN** to on.

## 4 Connect everything

**4a** Plug the cable's RJ45 plug into the CAN port you chose in 3c, with the VE.Can terminator in the port's other socket (as in step 2h).

**4b** Plug the D1's signal plug back in and switch its 12 V supply (KL30/KL15) back on.

> Nothing changes for the D1, now or later. The monitor can only listen: it has no way to send the D1 a command.

## 5 Install the monitor with Cerbo Setup

**5a** **Open the helper.** In the project folder, double-click `Cerbo Setup.bat`. If Windows shows "Windows protected your PC", click **More info** and then **Run anyway**.

**5b** **Type the Cerbo's IP address** from step 3b and press **Enter**. The helper remembers it for next time, and then shows its menu:

```
  ============================================================
    Nomadic DC-DC monitor  -  Cerbo GX setup
    Cerbo address: 192.168.1.50
    Monitor only: nothing is ever sent to the converter
  ============================================================

    1  Edit the settings (config.ini)
    2  Find which Cerbo port the converter is on
    3  Watch live readings from the converter
    4  Install or update the monitor on the Cerbo
    5  Show the monitor log

    9  Try it with a pretend converter (simulation)

    6  Change the Cerbo address
    7  Fix "REMOTE HOST IDENTIFICATION HAS CHANGED"
    8  Uninstall the monitor from the Cerbo
    0  Exit

  Type a number and press Enter: _
```

> **Want to see it first?** Option **9** runs everything with a pretend D1 and no CAN bus at all (see the Safe Test Plan, `docs/safe-test-plan.pdf` ).

> **The password prompt.** Every option that talks to the Cerbo asks for the root password from step 3d. **Nothing appears on screen while you type it.** That's normal: type it and press Enter.

**5c** **Type `2` : find the converter.** The helper copies the monitor to the Cerbo and listens on every CAN port for 3 seconds. You should see something like this:

```
Venus OS v3.80 OK
Listening 3 s on each CAN port (nothing is transmitted)...
  can0    250 kbit/s  error-active  nothing from the converter
  can1    250 kbit/s  error-active  CONVERTER FOUND (391 frames)
The converter is on can1, which matches config.ini. Good.
```

If it says "In config.ini set: interface = can0", make that change in the next step. If the D1 is set to the other speed, you'll see bus errors instead:

```
  can1    250 kbit/s  error-passive nothing decoded, but 288 bus errors: talking at a different
speed?
Something is talking on can1 at a different speed. On the Cerbo, set that port's CAN-bus
profile to "Up, but no services (500 kbit/s)" and run this again.
```

Do what it says (step 3c), then type `2` again. If nothing is heard at all, see Troubleshooting, and check the KL30/KL15 supply first.

**5d** **Type `1` : edit the settings.** Notepad opens `config.ini`. Check these lines, then press Ctrl+S to save and close Notepad:

| Line | What to put |
|---|---|
| `interface = can1` | The port that step 5c found |
| `custom_name = DC-DC D1` | The name shown on the Cerbo and in VRM |
| `password =` | Optional: a password for the web page. Without one, anyone on your network can see the readings (but can't change anything) |
| `switch_pane = yes` | Leave as yes to get the cards on the Cerbo screen and in VRM |

**5e** **Type `3` : watch the live readings** (optional, but a good check). The voltages should match what a meter shows on the D1's terminals. To stop, press Ctrl+C, then type N and press Enter.

```
--- can1 10:42:07  ONLINE  0x066=20/s 0x088=100/s 0x520=10/s
LV  13.4 V  +22 A   +295 W | HV  51.2 V   -6 A   -307 W | 35 C
Mode: Autonomous (default set-point) | Precharge not in process
Alarms: none
Software 1.2.3  Hardware 5
```

**5f** **Type `4` : install.** After about 10 seconds you should see:

```
/service/dbus-nomadic-dcdc: up (pid 1834) 6 seconds
... INFO: Nomadic DC-DC monitor 2.0.0 starting on can1 (monitor only: nothing is transmitted)
... INFO: Converter online on can1
```

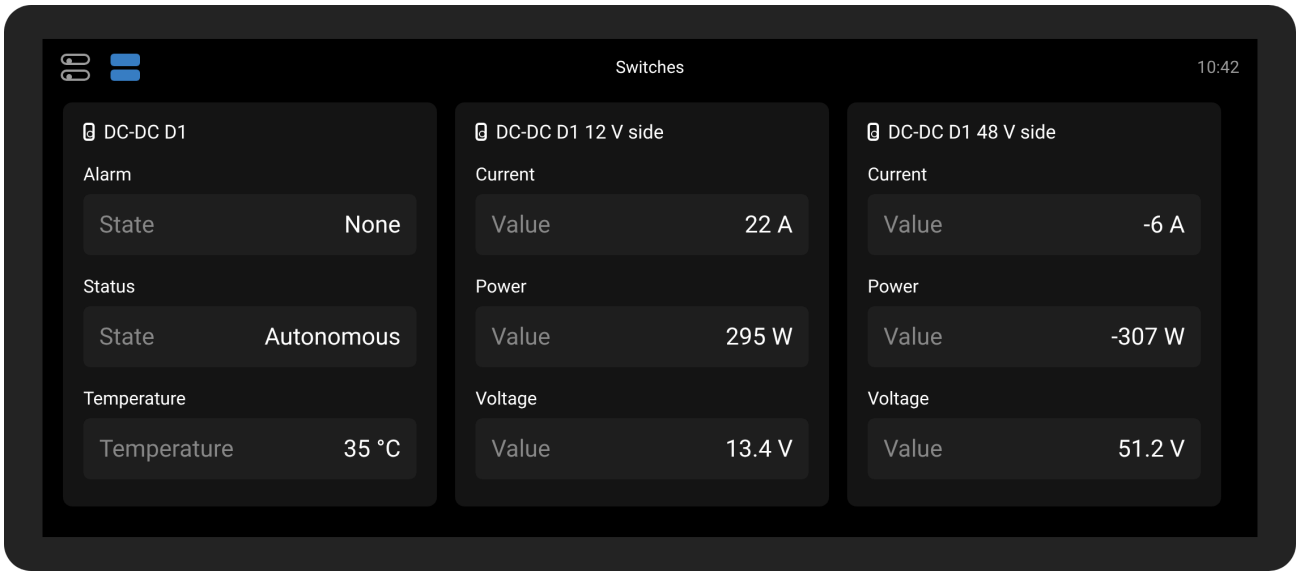
**Done!** It now starts by itself after every reboot and firmware update.

> **Upgrading from the old control version (1.x)?** Installing this monitor (option 4) replaces it. If the old version was in Off, Buck or Boost, its commands stop when the monitor starts, and nobody knows yet whether the D1 then goes back to its own logic or keeps the last command. So first switch the old version to **Automatic** (it then sends nothing), check the D1 behaves as it does on its own, and only then install. If it seems stuck, switch KL15 off and on to restart its controller.

## 6 Use it

### On the Cerbo screen (and Remote Console)

Tap the **switches icon** at the top left of the screen. Three cards appear (swipe if your screen shows fewer at once):




| Card | Item | Shows |
|---|---|---|
| **DC-DC D1** | Alarm | The most serious active alarm, or None. The tile turns red on a fault, and "N active" appears when there are several |
| | Status | What the D1 says it's doing: Standby, Autonomous, Buck, Boost, Derating (hot)… or **No communication** |
| | Temperature | The D1's power board temperature, in your Cerbo's temperature unit |
| **DC-DC D1 12 V side**<br>**DC-DC D1 48 V side** | Current, Power, Voltage | Live readings for that battery side. **+** means energy going into that battery, **−** coming out (see below) |

To rename, move or hide any item, go to **Settings › Device list › DC-DC D1**. A hidden **Precharge status** item is there too (the D1 manual doesn't use precharge). The D1 also appears in the **Device list** as a DC-DC converter, with its 12 V side as the output and its 48 V side as the input.

### In the VRM app and portal

The same cards appear in the **Switches** widget in the VRM app, and in the **Switches** panel on the VRM portal dashboard. The Cerbo must be online with VRM. Nothing needs to be changed in VRM's access settings: there is nothing to control.

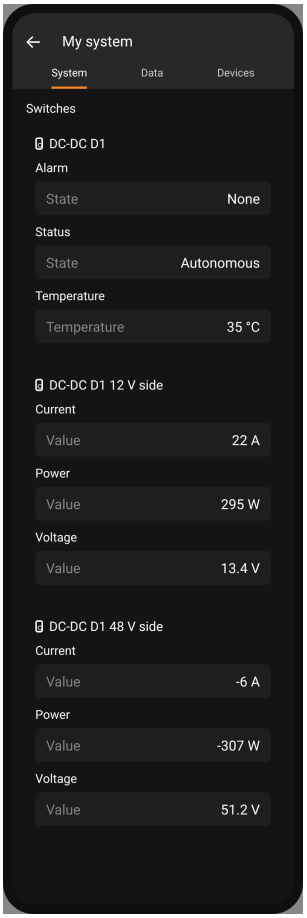
The picture is an illustration. VRM draws its own version of these cards, so the layout may differ a little.

### On a web page, with charts

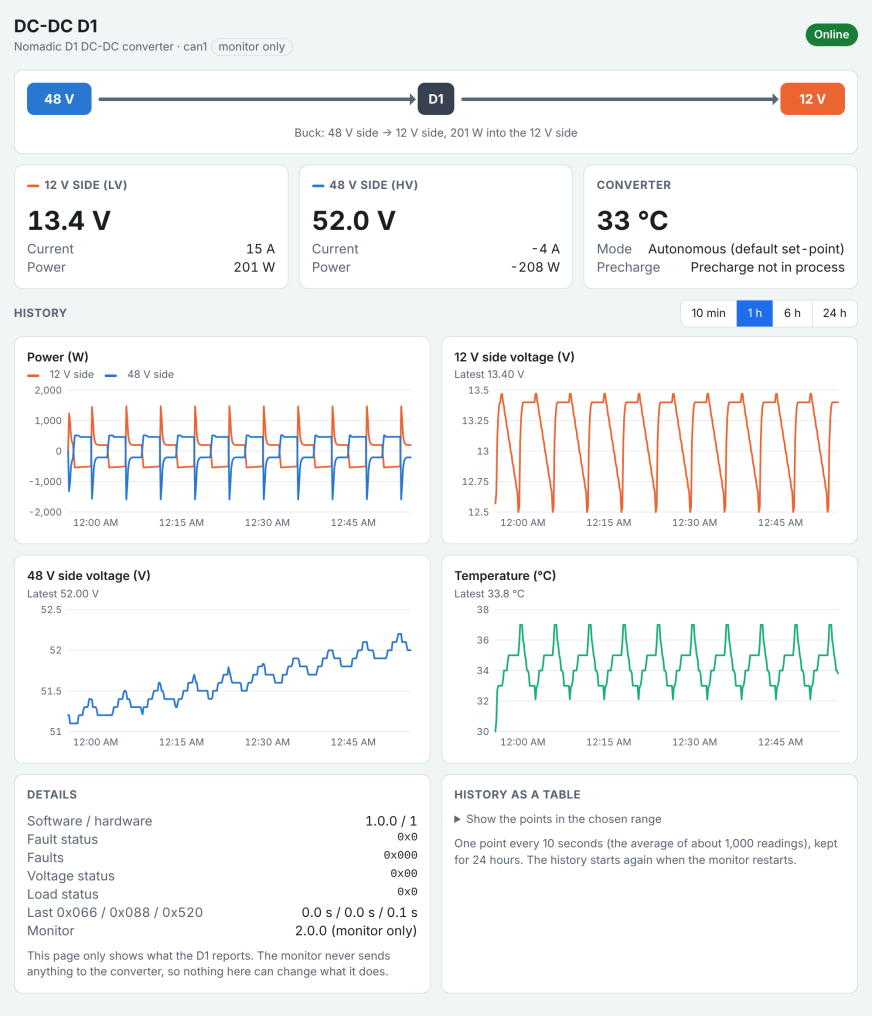
On any phone, tablet or computer on the same network, open:

## http://<Cerbo IP>:8088

It shows the live readings, which way the energy is flowing, every alarm, the raw status bits, and charts of the last 10 minutes to 24 hours. Point at a chart to read the values at that moment. The history starts again when the monitor restarts.

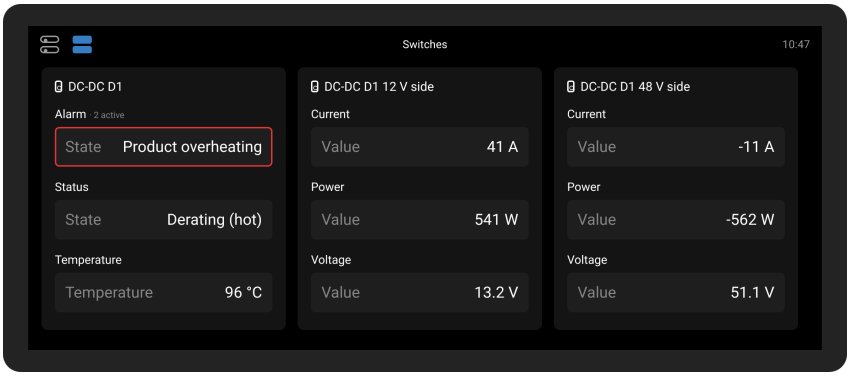



VRM app, Switches widget (illustration)




The web page, with a pretend D1 (simulation).

## When something is wrong




An example: the D1 is hot, so it's limiting its power (derating). The alarm shows the most serious problem first.

## Reading the numbers

### Which way is the energy going?

**BUCK: 48 V HELPS 12 V**

48 V ····· D1 ····· 12 V

12 V side current **+22 A** · power **+295 W**
48 V side current **−6 A** · power **−307 W**

**BOOST: 12 V HELPS 48 V**

48 V ····· D1 ····· 12 V

12 V side current **−40 A** · power **−520 W**
48 V side current **+9 A** · power **+470 W**

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**+** means energy is going **into** that battery, **−** means it's coming **out**. The side that's being charged is the one with the plus sign. The 48 V side always shows a little more than the 12 V side gets in buck (and the other way round in boost): that difference is the D1's own losses.

The D1's documents don't say which way its current is counted. The monitor shows exactly what the D1 reports; check the sign once with a clamp meter.

On its own, the D1 **bucks** (charges the 12 V side from 48 V) when the 12 V side drops below 12.5 V, as long as the 48 V side is 41–58 V, and fills it back up to about 13.4 V. It **boosts** (pushes 12 V energy up to 48 V) when the 12 V side is 13.5–16 V, until it falls below 12.3 V. In between it waits in **Standby**, with no current.

## ? Troubleshooting

| What you see | What to do |
|---|---|
| Option 2: "not heard on any port" | Check, in this order: (1) the D1 has KL30 (pin 4, 9–16 V) and KL15 (pin 3, 8 V or more), measured to GND. Without both, its controller is off and silent; (2) the CAN speed: if it reports bus errors, switch to the other speed (step 3c); (3) the wires (2b), and that CAN_H/CAN_L are in pins 1/2 of the D1 plug; (4) the termination reads about 60 Ω (2h); (5) the cable is in the Cerbo port you set up. |
| "ssh" or "tar" is not recognized | Install the Windows OpenSSH client: Settings › Apps › Optional features › Add a feature › OpenSSH Client. |
| "Connection timed out" or "Connection refused" | Check the IP address (menu option 6), that the PC and Cerbo are on the same network, and that Enable SSH on LAN is on (step 3d). |
| "Permission denied" | Wrong root password. Set it again (step 3d, item 3). |
| "REMOTE HOST IDENTIFICATION HAS CHANGED" | This happens after the Cerbo is reset or replaced. Use menu option 7, then try again. |
| Option 2: Venus OS version warning | Update the Cerbo (step 3a). Older firmware can't show the cards. |
| Nothing appears in the switch pane | Check the firmware is v3.80 or newer. Use option 5 (log) and look for "Registered ... on D-Bus". Make sure `switch_pane = yes` in config.ini. |
| Status shows Standby | Normal when the 12 V side is between 12.5 and 13.5 V: the D1 has nothing to do. |
| Status shows No communication | The Cerbo can't hear the D1. Check KL30/KL15, the cable and the termination. |
| The charts say "Collecting data..." | They add a point every 10 seconds. Wait a minute. They start empty after every restart of the monitor or the Cerbo. |
| The "DC system" figure on the Cerbo looks wrong | The Cerbo counts the D1 as a DC-DC charger. Set `service_class = nomadicdcdc` in config.ini (option 1, then 4). The D1 leaves the device list, but the cards, VRM and web page keep working. |

**To change settings later,** open Cerbo Setup, choose 1 to edit config.ini, save, then choose 4. **To remove it,** choose 8 and type YES. The D1 is not affected either way.